

STAKEHOLDER AND PROBLEM ANALYSIS

1. Stakeholder and Organization

Stakeholder: Kgomotso Ranchu

Organization/Business: Agricastadvisory

The primary stakeholder is involved in the commercial livestock sector, where livestock owners/sellers need access to buyers and suitable markets for their animals. The stakeholder identified the need for a digital solution that can improve the way livestock, crops and feed is presented to potential buyers and make it easier for sellers and buyers to connect.

The proposed system is intended to support both livestock buyers and sellers, with the main focus being on improving access to the livestock market.

Current Process

Based on the stakeholder engagement, livestock sellers currently need to manage several activities when trying to bring their livestock to market.

A livestock owner needs to have information about their animals available, including information such as:

- The type of livestock.
- The animal's weight.
- The animal's branding/ownership information.
- When the animal was born.
- The animal's parents.
- Vaccination information.
- Documentation that can demonstrate ownership of the livestock.

The stakeholder also identified the importance of being able to reach livestock from different farms and connect with potential buyers.

The current market process can therefore be broadly represented as:

Livestock owner → Identify livestock → Record/prepare livestock information → Find potential market/buyer → Negotiate/sell livestock

The stakeholder indicated that opportunities such as **auctions** and **feedlots** are relevant to the commercial livestock market. The proposed system would provide a digital way for buyers and sellers to access livestock-related market opportunities.

Current-process problem

The current process creates a challenge when livestock owners need to reach suitable markets and potential buyers. Information about individual livestock may also need to be recorded and presented when the animal is being marketed.

This creates an opportunity for a centralized system where livestock sellers can present their livestock and relevant information to potential buyers.

People Involved / Stakeholders

The proposed system will involve several categories of users.

Primary users

1.1 Livestock Sellers/Farmer

Livestock sellers will use the system to:

- Record livestock information.
- Provide information such as weight and branding.
- Upload documentation proving ownership.
- Advertise livestock for sale.
- Specify a price where applicable.
- Make their livestock visible to potential buyers.

1.2 Livestock Buyers

Buyers will use the system to:

- View livestock available from different farms.

- Access relevant livestock information.
- Identify livestock that meets their requirements.
- Contact/connect with sellers.
- Access available market opportunities.

Other relevant participants

The stakeholder also specified **auctions and feedlots** as relevant parts of the livestock market that will use the prototype. These also form part of the wider ecosystem supported by the proposed system.

2. Problem Analysis

The primary problem identified by the stakeholder is **limited and inefficient access to the livestock market for commercial livestock sellers and buyers**.

Livestock sellers need effective ways to make their animals visible to potential buyers and provide vital information about the livestock. Buyers, on the other hand, need a way to discover livestock from different farms and obtain relevant information before deciding whether to purchase.

The stakeholder therefore showed a need for a system that can **connect livestock buyers and sellers through a centralized digital platform**, while providing information about the livestock being offered.

The problem is particularly focused on **market access**, rather than simply livestock record keeping.

Consequences of the Problem

The identified problem can result in:

- Livestock sellers having difficulty reaching a wider pool of potential buyers.
- Buyers having difficulty discovering livestock available from different farms.
- Livestock information being difficult to access when evaluating an animal.
- Sellers potentially relying on fragmented methods to advertise or communicate livestock availability.
- Difficulty connecting livestock with suitable commercial market opportunities (farmers struggle to find buyers).
- Reduced visibility of livestock available for sale (buyers do not know where to buy).

Users affected by the problem

Affected user	How they are affected
Livestock sellers/farmers	May struggle to access a wider market and present their livestock information to potential buyers.
Livestock buyers	May have difficulty finding livestock from different farms and accessing information needed to make purchasing decisions.
Commercial livestock market participants	May experience difficulties connecting livestock with suitable market opportunities.

Problem Statement

Commercial livestock and crop sellers currently face challenges in accessing suitable markets and reaching potential buyers, while buyers may struggle to discover livestock from different farms and access reliable information about the animals. The livestock business and local farmers primarily rely on manual paper registers to maintain animal records and physical auctions to facilitate sales. This fragmented approach can result in limited market reach, difficulties in tracking animal health and ownership information, and increased risks associated with the loss or misuse of paper-based ownership documentation. The problem affects farmers, buyers, feedlot managers and traders, who may need to be physically present to access records, negotiate transactions and verify ownership documentation.

Therefore, there is a need for an integrated digital marketplace and livestock record system that centralises relevant information and improves access to commercial markets. The proposed solution will enable sellers to present livestock, crops and animal feed to a wider pool of potential buyers while allowing relevant livestock information, including the breed, health records, weight and digital brandmark documentation all in one platform. By connecting buyers and sellers digitally and reducing reliance on paper-based records and exclusively physical trading processes, the system aims to improve market visibility, simplify transactions and provide a more accessible and efficient approach to commercial livestock trading.